ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25
B.Tech. (CSE)
Semester: IV

SUBJECT CODE: BCS403
OPERATING SYSTEMS

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Answer the following questions: (1 × 5 = 5 Marks)

(i) Define a real-time operating system.
(ii) What are the advantages of demand paging?
(iii) Explain race condition with example.
(iv) Differentiate between user-level and kernel-level threads.
(v) Explain the concept of context switching.

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) What are the advantages and disadvantages of multiprogramming?
(ii) Explain the concept of non-preemptive vs. preemptive scheduling.
(iii) Define semaphore and its role in process synchronization.
(iv) What is fragmentation? How does paging help in reducing fragmentation?
(v) Differentiate between deadlock and starvation.

SECTION – B

NOTE:Attempt all questions. **(5×2×2 = 20 Marks)**

Q.3.(i)(a) Consider a cache memory system where 1000 memory accesses are made. If 750 of these accesses result in cache hits, calculate the hit ratio and miss ratio. Also, discuss how cache performance is affected by these ratios.

OR

(**b)** Explain deadlock prevention vs. deadlock avoidance with suitable examples.

(ii)(a) Implement Preemptive Priority scheduling for processes P1, P2, P3, P4, and P5, given the following details:

Process	Arrival Time	Burst Time	Priority
P1	0	5	2
P2	1	3	1
P3	2	8	4
P4	3	6	3
P5	4	2	5

Find the average waiting time, average turnaround time and throughput. (Higher value higher priority).

OR

(b) Explain the differences between multiprocessor and clustered systems in operating systems.

Q.4.(i)(a) What is a Critical Section? Explain Peterson's Solution for the critical section problem.

OR

(**b)** What are the disadvantages of segmentation memory allocation? Explain with an example.

(ii)(a) What is Virtual Memory? Explain the working of the Page Table in the context of virtual memory.

OR

(**b)** A disk with 200 tracks (0-199) has the following pending requests at some time: 82, 170, 43, 140, 24, 16, and 190. The current head position is 50, and the previous request was 45. Compute the total head movement using LOOK and C-LOOK algorithms.

SECTION – C

NOTE: Attempt all questions. **(10 × 2 = 20 Marks)**

Q.5.(a) Why is Page Replacement necessary? Consider the following page reference string: **2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6** Calculate the number of page faults using the LRU (Least Recently Used) page replacement algorithm, assuming three frames and initially empty frames.

OR

(b) Explain Banker's Algorithm for deadlock avoidance. Solve an example problem where a system has five processes (P1-P5) and four resource types (A, B, C, D) with available resources (3, 3, 2, 2). Given the maximum need and allocation matrix, determine whether the system is in a safe state.

Process	Allocation				Max_Need			
	A	B	C	D	A	B	C	D
P1	0	1	0	1	7	5	3	1
P2	2	0	0	1	3	2	2	2
P3	3	0	2	1	9	0	2	2
P4	2	1	1	0	2	2	2	2
P5	0	0	2	2	4	3	3	3

Q.6.(a) What do you mean by RAID structure? Explain RAID levels 0, 1, 5, and 6 with their advantages and disadvantages.

OR

(b) Write notes on:

1) Thrashing.
2) Page fault handling.